

Composite Armor in Combat Robotics Applications

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1 Abstract

In the field of combat robotics across most weight divisions, AR500 steel is often chosen due to its high impact resistance. However, where weight minimization is a large factor in design, this could be mitigated by the choice of various composite laminate materials to match the impact-energy absorption of AR500 steel. We aim to explore these impacts in an FEM Ansys LS-DYNA simulation by establishing a baseline test of energy absorption to a fixed volume steel plate. We will then investigate various carbon fiber reinforced polymer (CFRP) composites of various ply orientations and layer counts to simulate under the control conditions. The energy absorption, as Joules per g, will be compared to each material in order to identify the idealized fiber orientation layup that can best absorb energy. We aim to identify a CFRP layup that can provide a practical strength-to-weight ratio within impact-prone, mass-constrained environments.

2 Introduction

Combat robotics features a variety of robot designs in an arena. Several notable designs are those of the horizontal and vertical spinner as seen in Figure 1. These feature a flat spinning bar or disk that is oriented in its respective axis to transfer kinetic energy and deliver damage to the opponent. Therefore, armor design is important to absorb these impacts to prevent core damage to the robot. The most common materials utilized for combat robotics armor are steel, aluminum, and plastic.

Armor is important in being able to absorb as much impact energy into its own structure rather than fracturing and allowing the weapon to transfer its kinetic energy into critical subsystems. For example, Bernal et al. showed that heat treatment of low-carbon steel increased the Charpy impact toughness from 97 J to 159 J when tested using ASTM E23 [2]. In comparison, Yildız reported that 6061-T6 aluminum had a Charpy impact energy of 50.3 J [3]. This demonstrates why steel is often used as a strong baseline armor material, while CFRP is analyzed in this study as a possible lightweight alternative for absorbing impacts.



Figure 1: Vertical Dual Disk Spinner - Escl Designed and built by RoboJackets Members (2017-2018)

3 Background

Robot combat impacts differ significantly from standard material testing conditions. Existing impact research on composites features those of low-velocity impacts (drop-weight tests) [4], high-velocity impacts (ballistics) [5], or simplified flat laminate models. As a result, there is limited research on composite performance under the impacts relevant to combat robotics. Existing design resources for combat robotics also lack a detailed analysis of composite configurations, leaving a gap in understanding how these materials could be effectively applied.

4 Objective

The selection of AR500 steel was utilized as a baseline material to get a baseline energy absorption and material state response to the impact. The CFRP layup orders will be compared against this control. The goal of this research is to identify a layup order in which the CFRP can compare to the strong, impact performance of steel, whilst providing an advantage in weight. Such an advantage can be exploited in a future robot design, allowing for more weight allocated to other subsystems. A hypothesis was made based on a study on the fatigue load-

ing of fiber reinforced polymer which identified a Quasi-Isotropic layup $[0/-45/45/90]$ as a potentially commonly used, efficient fiber stackup. We expected the absorption energy to increase and the impactor kinetic energy to decrease proportionally as the number of layers went up

5 Methods

5.1 Simulation Workflow

The study compares the impact response of an AR500 steel armor plate to CFRP laminate configurations. The impact condition is kept constant for every armor configuration for consistency. Due to its availability to the market and to maintain an industry-standard cross-section for an impactor, the Repeat Robotics Beater Bar was utilized, which is meant for the 3lb weight class [8]. The setup is shown in Figure 3 using Ansys LS-DYNA. The plate is a simplified model of the armor equipped on a standard 3lb combat robot. The boundary conditions are fixed on the left and right sides of the plate.

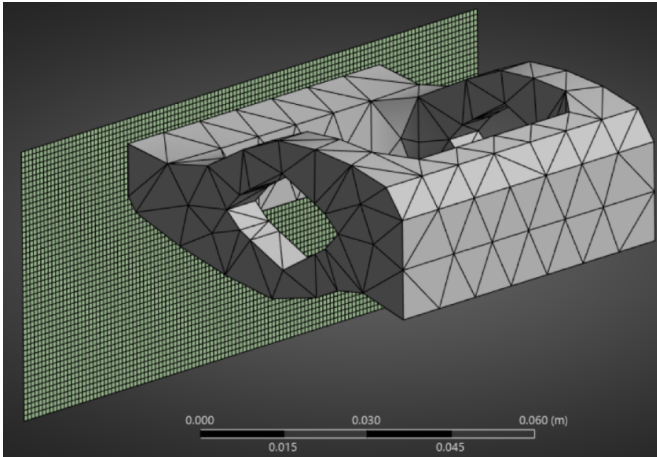


Figure 2: Impact Setup in ANSYS Mechanical

Table 1 shows the variables that are kept constant in the study for all armor configurations between both the AR500 control and CFRP layup tests. As there is not a specific standardized process for this test in particular, the beater-bar tip speed was set to 250 mph because this value represents a common upper-limit weapon speed used in combat robotics rulesets as described in BYU combat robotics [9] and in the Southern California Attack Robotics 2025 rule-set [10]. The translational velocity toward armor plate is a modeling assumption as there is no common speed demonstrated by robots in the arena, though 10 mph is generally accepted to be on the higher end of drive capabilities. The impact angle represents the angular deviation of the impactor axis of rotation orthogonal from the plate. As such, this study will only look at the impactor striking the plate with no change of angle. The bite depth is another modeling assumption to model the striking location of the impactor;

the value chosen allows the impactor to make a full rotation before impact, resulting in maximum kinetic energy transfer. The plate length and width are all kept constant whilst the AR500 plate thickness and mass are characterized to be compared to in the CFRP parameters. The plate dimensions were decided as to be visually consistent with the dimensions of most robots within the 3 lb weight class.

As different manufacturers cite different material properties for AR500, it was decided to utilize the data from SendCutSend [11], with the properties utilized for simulation shown in Table 2.

The CFRP laminate studies follow the primary laminate families as shown in Table 3. The two unidirectional laminates evaluate the fiber direction’s influence to the impact response. The various cross-ply, angle-ply, and quasi-isotropic layups were analyzed to determine how the response would be with varying angle layers.

Table 2: AR500 material properties, converted to SI units.

Material Property	Value
Young’s Modulus	207 GPa
Poisson’s Ratio	0.29
Bulk Modulus	164 GPa
Shear Modulus	80.2 GPa

Table 3: Laminate families and layup orders used in the simulation matrix.

Laminate Family	Layup Order
Unidirectional 0°	$[0]_n$
Unidirectional 90°	$[90]_n$
Cross-ply	$[0/90]_n$
Angle-ply	$[+45/-45]_n$
Quasi-isotropic	$[0/+45/-45/90]_n$

5.2 Analysis Workflow

The main purpose of comparing the energy absorption between the AR500 steel and CFRP laminates comes from mapping the outputs of maximum internal energy, kinetic energy, total deformation, max deformation, equivalent max stress, and equivalent average stress of the system. The deformation and stress are also analyzed in order to identify useful deformation or damage to the system.

Energy absorption and specific energy absorption (*SEA*) are studied in this paper because they are widely used in impact studies for composite laminates [12, 13]. They often analyze the overall absorbed energy, contact response, damage extent, and stacking sequence [12, 13], which is what we aim to achieve in the CFRP analyses to the AR500 steel. Since combat robotics are within mass constrained restrictions, we can use specific energy absorption to reflect that similar to existing com-

Table 1: Constant simulation parameters used for all armor configurations.

Parameter	Value	Units
Impactor type	Repeat Robotics Beater Bar	–
Impact event	Single impact cycle	–
Beater-bar tip speed	250	mph
Beater-bar diameter	75	mm
Translational velocity toward armor plate	10	mph
Angle of armor plate	0	deg
Bite depth	9.4	mm
Plate length	125	mm
Plate width	50	mm
AR500 plate thickness	3.023	mm
AR500 plate mass	150.616	g
Varied CFRP parameters	Layup sequence and number of plies	–

posite studies and identify performance in varying composites [12, 13].

In order to establish a baseline, the absorbed energy is identified as the internal energy accumulated in the AR500 plate during the impact event:

$$E_{\text{absorbed, steel}} = E_{\text{internal, steel}} \quad (1)$$

where $E_{\text{absorbed, steel}}$ is the absorbed impact energy of the AR500 control and $E_{\text{internal, steel}}$ is the internal energy of the steel plate extracted from the finite element simulation. The laminate configurations must exceed or match this absorbed impact energy. Therefore, a certain laminate configuration was considered to match the energy absorption of the steel control if it is shown that its absorbed energy is equal to or greater than the absorbed energy of steel.

$$E_{\text{absorbed, laminate}} \geq E_{\text{absorbed, steel}} \quad (2)$$

where $E_{\text{absorbed, laminate}}$ is the absorbed impact energy of the CFRP laminate. This criterion does not by itself determine the best armor configuration; it only indicates whether the laminate absorbed at least as much impact energy as the steel control.

The overall weight efficiency is also characterized to analyze how the energy absorption performs in terms of unit mass from the specific energy absorption as represented as SEA :

$$SEA = \frac{E_{\text{absorbed}}}{m} \quad (3)$$

where SEA is the specific energy absorption, E_{absorbed} is the absorbed impact energy, and m is the mass of the armor plate. This helps to determine whether a laminate absorb the energy properly at lower mass.

Dimensionless ratios were utilized in order to create normalized scale to the AR500 control. The first ratio is the energy absorption ratio that compares the energy absorbed to the control.

$$R_E = \frac{E_{\text{absorbed, laminate}}}{E_{\text{absorbed, steel}}} \quad (4)$$

where R_E is the energy absorption ratio. A value of $R_E < 1$ indicates that the laminate absorbed less energy than the steel control, $R_E = 1$ indicates that the laminate matched the steel control, and $R_E > 1$ indicates that the laminate exceeded the steel control in absorbed energy.

The second dimensionless ratio is the mass ratio, described as:

$$R_m = \frac{m_{\text{laminate}}}{m_{\text{steel}}} \quad (5)$$

where R_m is the mass ratio, m_{laminate} is the laminate mass, and m_{steel} is the mass of the AR500 control. A value of $R_m < 1$ indicates that the laminate is lighter than the steel control, while $R_m > 1$ indicates that it is heavier.

To identify the best laminate configuration, both the energy and mass ratio should be satisfied from the following criteria:

$$R_E \geq 1 \quad \text{and} \quad R_m < 1 \quad (6)$$

This would show that the laminate has absorbed as much energy as the AR500 steel whilst having lower mass. In the event that none reach this configuration, the next best score that reaches close to the criterion will be analyzed.

5.3 Control Model

The Johnson-Cook constitutive model was utilized to better calibrate the AR500 steel control as it is often utilized under larger strain or high strain rate simulations [7]. The Johnson-Cook model is labeled as the following:

$$\sigma_y = (A + B\varepsilon_p^n) (1 + C \ln \dot{\varepsilon}^*) (1 - T^{*m}) \quad (7)$$

where σ_y is the flow stress, A is the initial yield strength, B is the strain-hardening coefficient, n is strain-hardening exponent, ϵ_p is equivalent plastic strain, C is strain-rate sensitivity coefficient, $\dot{\epsilon}^*$ is the normalized plastic strain rate, T^* is the homologous temperature, and m is the thermal softening exponent.

The normalized strain rate is defined as:

$$\dot{\epsilon}^* = \frac{\dot{\epsilon}_p}{\dot{\epsilon}_0} \quad (8)$$

where $\dot{\epsilon}_p$ is the equivalent plastic strain, and $\dot{\epsilon}_0$ is the reference strain rate.

The homologous temperature is represented as:

$$T^* = \frac{T - T_0}{T_m - T_0} \quad (9)$$

where T is the current material temperature, T_0 is reference temperature, and T_m is the melting temperature. In our case, we assume the current and reference temperatures are equivalent since temperature is very rarely a deciding factor in combat robotics.

Overall, the equation is threefold. First, it shows how the steel strengthens as it plastically deforms in $(A + B\epsilon_p^n)$. The overall impact could create local plastic deformations only close to that of the contact region, and the strain hardening would further calibrate the simulation. Second, the flow stress is shown to increase during larger strain rates in $(1 + C \ln \dot{\epsilon}^*)$. This is extremely impactful in our current study, where we are driving a high velocity object to impact the plate, yielding an extremely high strain rate in a small amount of time. Third, thermal softening would cause less stress loading as represented in $(1 - T^{*m})$.

Applying the Johnson-Cook strength equation will model the flow stress, but there is no explicit crack formation or material separation models utilized, so limitations for this control will only be representative of the plastic deformation and energy absorption, and not that of identifying a proper failure mode. Additionally, further limitations of these are due to identifying approximations received in papers from Forrestal et al. [5] and Manes et al. [6].

The following Johnson-Cook parameters were taken from a ballistic study featuring various steels and aluminum [1] and were implemented into the material on Ansys.

Table 4: Johnson-Cook material constants used for the AR500 steel control.

Material Constant	Value	Units
A	1250	MPa
B	362	MPa
n	1	Unitless
C	0.0108	Unitless
m	1	Unitless
T_m	1800	K

5.4 CFRP Model

The CFRP laminate was modeled differently than the AR500 steel due to its orthotropic properties. Epoxy Carbon UD (230 GPa) Prepreg with MAT054 is selected from among composite models in the built-in ANSYS Composite Materials database. MAT054 was chosen as the CFRP composite material model in LS-DYNA due to its reasonable accuracy and simpler model than compared to MAT058 and MAT262.

For a brief comparison, MAT054 assumes linear elastic orthotropic behavior with no pre or post-peak softening and couples transverse matrix compressive failure and longitudinal fiber compressive strength. On the other hand, MAT058 incorporates non-linear pre and post-peak softening and does not couple the compressive strengths, causing peak forces to be overpredicted [15]. Finally, MAT262 is the most rigorous of the three models but requires a very specific calibration process to tune inter-linked parameters with a specific Design of Experiment, which is far too complicated and unnecessary in our case. For these reasons, we chose to stick with MAT054 for simplicity, using literature parameters, which is common for materials without experimental data.

The orthotropic ply properties required by the CFRP model are summarized in Table 5. E_A is the Young's modulus in the longitudinal fiber direction, E_B is the Young's modulus in the transverse direction, G_{AB} and G_{BC} are shear moduli, and X_T , X_C , Y_T , Y_C , and S_L represent tensile, compressive, and shear strengths respectively.

Table 5: Orthotropic ply properties required for CFRP material modeling.

Property	Value	Units
ρ	1.58×10^{-6}	kg/mm ³
E_A	165000	MPa
E_B	9000	MPa
ν_{BA}, ν_{CA}	0.0185	Unitless
ν_{CB}	0.5	Unitless
G_{AB}, G_{CA}	5600	MPa
G_{BC}	2800	MPa
X_C	1590	MPa
X_T	2560	MPa
Y_C	185	MPa
Y_T	73	MPa
S_L	90	MPa

The MAT054 parameters are used to characterize damage and erosion of the CFRP to compensate for the failed elements due to load-carrying ability 6. These parameters are utilized to control the post-failure behavior of node elements. The MAT054 damage parameters were selected based on prior LS-DYNA CFRP crush modeling work [15].

Table 6: MAT054 parameters used for CFRP damage behavior.

Parameter	Value
TFAIL	1.0×10^{-7} s
EPS	0.55
SOFT	0.57
SOFT2	—
PFL	100
BETA	0.00
SLIMT1	0.01
SLIMC1	1.00
SLIMT2	0.10
SLIMC2	1.00
SLIMS	1.00
FBRT	0.00
YCFAC	2.00

TFAIL is related to element deletion based on the stable time step, EPS defines an effective failure strain, and SOFT controls the reduction in strength in damaged crashfront regions. The SLIM parameters define the remaining stress-carrying ability after different failure modes, such as fiber tension, fiber compression, matrix tension, matrix compression, and shear failure.

6 Ansys LS-DYNA Setup

The armor plate begins 1 mm away from the flat face of the weapon, allowing for the weapon to achieve maximum “bite”, which is colloquially known as the depth a spinning weapon tooth “penetrates” an opponent’s armor during a collision. Maximum bite results in maximum transfer of kinetic energy to the plate. The bottom of the plate and the bottom of the weapon’s path are at the same height. Any increase in thickness occurs outward relative to the weapon’s origin. The initial CAD design can be seen in Figure 3.

6.1 Workbench

The entire workflow of the simulations is generally shown in Figure 4. The workflow starts with defining the material properties within the engineering data of AR500 steel and CFRP laminates. On the AR500 steel model, the isotropic material properties and the Johnson-Cook damage parameters were simply applied to the LS-DYNA model from Figure 4. Then, the simulation was ran given the control variables in Figure 1. Second, the CFRP laminates were first provided their orthotropic ply properties and MAT054 parameters in Figure 5 and Figure 6. Then, they are processed through ACP (Pre) for the customized set of orientation and layer numbers given in Figure 3. These parameters are ran through the same control variables from Figure 1 in LS-DYNA. A parameter set was overall linked in the entire workflow to provide a stream-

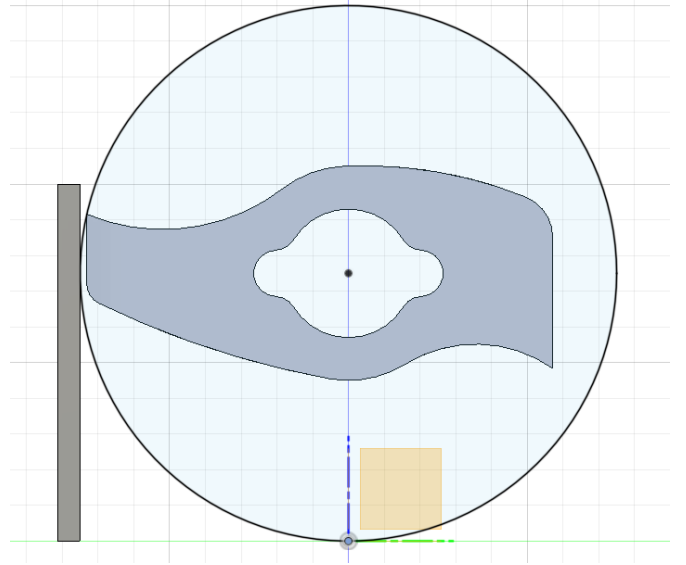


Figure 3: Initial Configuration in Fusion 360

lined comparison between all of the varying armor configurations for CFRP. This enabled the automation of running many of the test cases in a single impact test setup without the need to manually adjust the parameters and receive a bulk output metric.

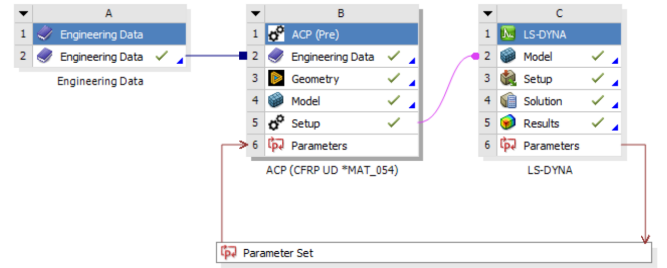


Figure 4: Ansys Workbench Flow

6.2 Geometry Model

The weapon impact model consists of a solid body impactor of the Repeat Robotics Beater Bar [8] with the armor plate of varying material parameters. The constants are defined in Figure 1, where this results in impactor tip distance starting 1mm away from the plate to allow for one full rotation before impact. Table 7 shows the mesh settings that were applied to the plate. This was chosen as a middle ground to balance the computation time and accuracy of the simulation. Since the plate behavior was the focus of our research, the impactor had a rough mesh of element size of 10mm and was set to have a rigid body for its stiffness behavior, for faster computational times. Our plate used a finer mesh with element sizes of 1mm. Analysis of meshing quality revealed an average of .97745 and standard deviation of 9.261e-002 under

Mesh Metric	Element Quality
Min	0.30057
Max	1.
Average	0.97745
Standard Deviation	9.261e-002

Table 7: Mesh Settings of Models in Simulation

the assumption that the idealized element shape is linear quad dominant (Square-shaped). A total element count of 6250 was for the plate, and 480 for the impactor.

The end time of the simulation was defined as 2.5e-3 seconds with a time step of 1e-7 seconds.

6.3 ACP

The 2D plate surface is used in ACP to build layers. Each ply is defined to have the default Epoxy Carbon Unidirectional material mentioned previously, with a thickness of 0.125mm. Given the plate's simple geometry, we used the default Globally Oriented Selection Set, which defined the layer stacking direction and the direction from which fiber orientations are defined. The stacking direction was purposefully set to be in the negative y-axis direction with the goal of stacking composite layers away from the impactor model. This ensures the plate remains the same distance from the impactor, securing the model's continuity with simulations of varying layer counts.

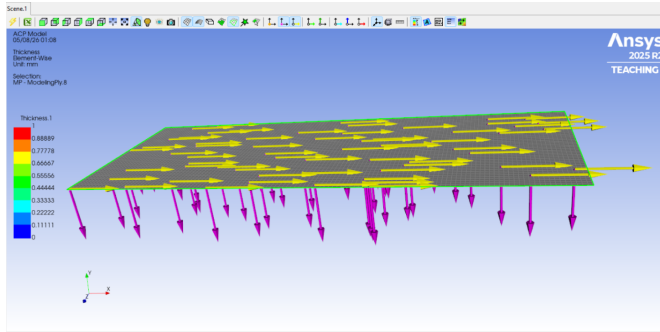


Figure 5: ACP principal directions. Purple: ply stacking direction in -y direction; Yellow: 0 degree fiber orientation direction.

These CFRP laminates were prepared for the setup by first creating 20 plies for the modeling group, which is the max number of plies tested. Second, each ply was given its own individual parameters for fiber orientation angle and whether the ply is suppressed or unsuppressed. With a total of 40 defined parameters, simulations were able to be run continuously while storing the visual data from the results of different fiber orientations and plate thicknesses for future analysis.

6.4 LS-DYNA

The impactor was defined with the initial conditions of 4.4704 m/s and 2980 rad/s for its linear and clockwise angular velocities, which were derived from the tip speed and translation velocity defined from Figure 1. Boundary conditions consisted of fixed edge boundary conditions along the z-oriented edges of the plate. Setting up the target solution results consisted of the following: minimum kinetic energy of the weapon, maximum internal energy of the plate, max and average deformation of the plate, and max and average stress of the plate.

6.5 Solution Plots

Finally, in order to capture the desired outputs for the analysis, the kinetic energy of the impactor, internal energy of the plate, total deformation, and equivalent stresses are all observed in the results of the simulations in Figure 6.

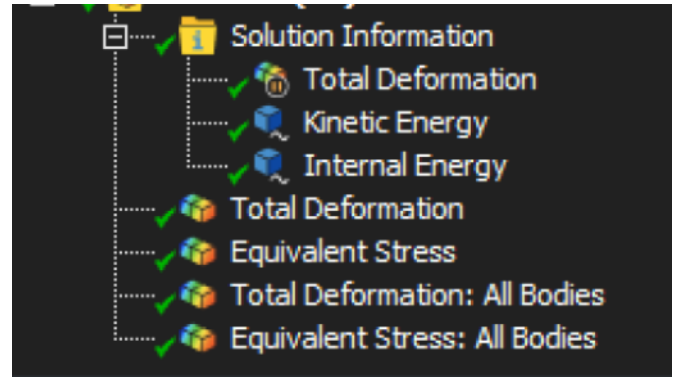


Figure 6: LS-DYNA Solution Plots

6.6 Low-velocity Beam Impact Setup

To compare the results of the plate-impactor simulation to other impact cases, simulations on non-notched laminates were also conducted. A beam geometry inspired by Charpy impact testing was used with the same laminate stack-ups as the plate simulations. For each case, the primary data collected was the energy absorbed during impact in order to compare how the different layouts responded to loading.

The setup is shown in Figure 7. A low velocity impactor weighing 0.5 kg impacted the beam with a velocity of 5 m/s. The impactor and rectangular supports were both modeled as rigid bodies. Boundary conditions included fixing the two supports while allowing the beam to deform freely under impact loading. The impactor and supports were meshed using 2 mm linear quad-dominant elements, while the beam itself was meshed using 1 mm elements. The Workbench workflow, ACP composite setup for the beam, and data collection process were performed in the same manner as the plate-impactor simulations.

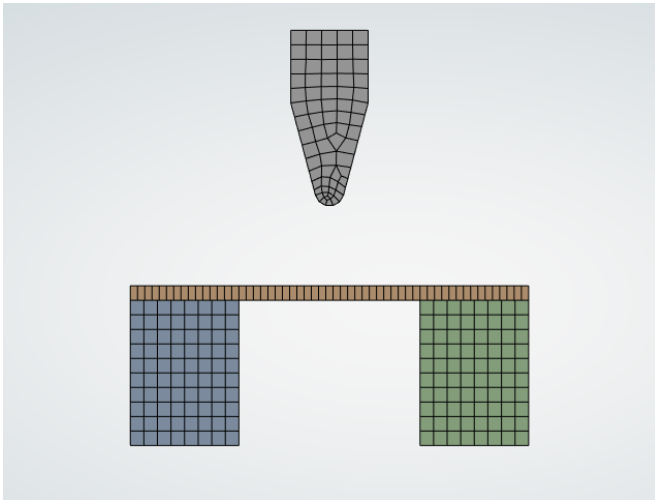


Figure 7: Low-Velocity Beam Impact Setup Geometry

7 Results and Analysis

7.1 Control

Before simulating composite performance, an AR500 steel plate was simulated as the baseline, since it is a common material for combat robotics armor. The results of the simulation are shown in Table 8. These results are the baseline of which we judge the performance of a composite laminate.

Metric	Value
Mass (g)	150.6
Thickness (mm)	3.02
Internal Energy Max (J)	302.56
Specific Energy Absorption (J/g)	2.43
Total Average Deformation (mm)	1.94

Table 8: Control Simulation Results

7.2 CFRP Simulation Results

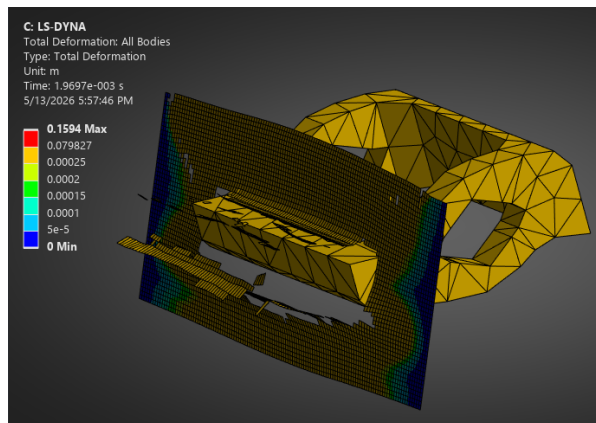


Figure 8: Unidirectional [0/+45/-45/90] 20 layer Deformation Simulation

Figure 9 shows the maximum internal energy of the 5 different composite laminate structures varied across the number of plies in the laminate. Internal energy effectively models the toughness of the armor in terms of how much energy it absorbs from the weapon. From the graph, it is clear that increasing the number of plies increases the internal energy of the plate. It seems that the cross-ply and angle plies generally produce slightly higher energy absorption than the others, though they are all rather similar in maximum internal energy.

However, while it was initially believed to be a good metric to consider, it was discovered to have inaccuracies. This is a result of the current setup, which involves elements being deleted after a certain strain limit, which then artificially lowers the measured internal energy. The simulation simplification was required, as the plies initially acted more akin to the fabric of clothing rather than carbon fiber. Regardless, a general trend can be observed.

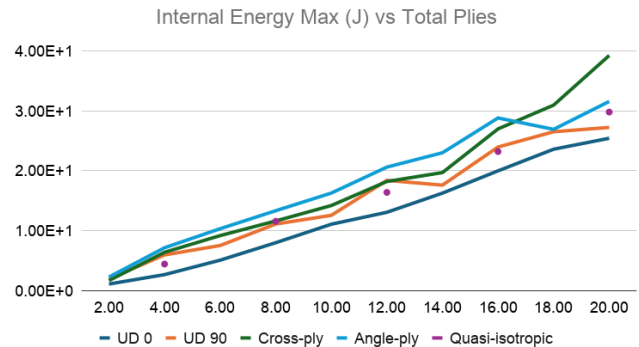


Figure 9: Internal Energy Max (J) vs Total Plies

Although high energy absorption is a critical metric for measuring combat robotics armor performance, it must also be structurally rigid in terms of deformation, since a high deformation can induce damage to underlying components. From Figure 10, it appears that the number of plies does not significantly affect the total average deformation of the armor, and overall, the deformation of the composites are significantly higher than the 1.94 mm deformation from the AR500 steel control armor.

Analysis of the deformation plots yielded results that were inconclusive. The deformation values remained fairly consistent, varying from unexpectedly large values of 20 mm to 30 mm. One reason for this could be that the fracturing and detaching of elements within the plate's mesh upon impact caused elements to be flung unrealistically far. Additionally the simulation was run at a very shortened time scale to keep computational times low, but as a result, kept total deformation relatively consistent.

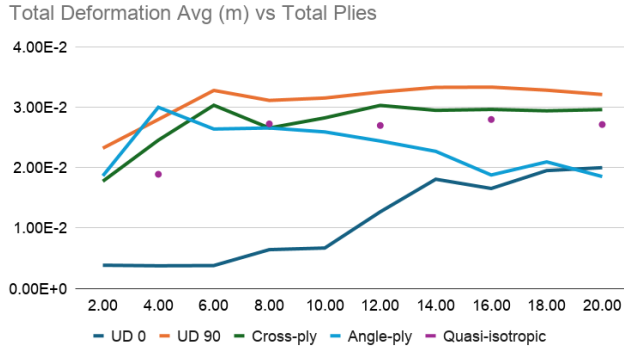


Figure 10: Total Deformation Avg (m) vs Total Plies

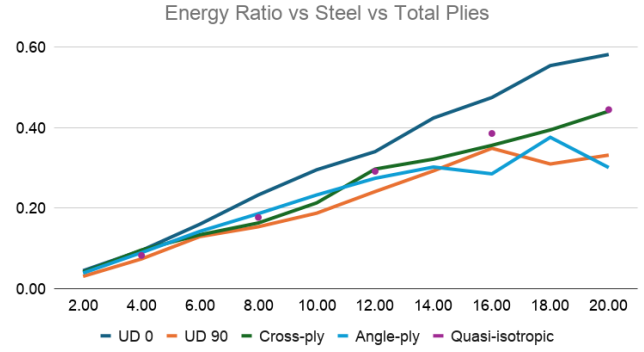


Figure 12: Energy Ratio vs Steel vs Total Plies

For a more apples-to-apples comparison, we compare energy absorption to weight for the different composites. Figure 11 shows the energy absorption per gram for all the different kinds of composites. The roughly flat line indicates that additional plies roughly increase the energy absorption roughly linearly, rather than exponentially.

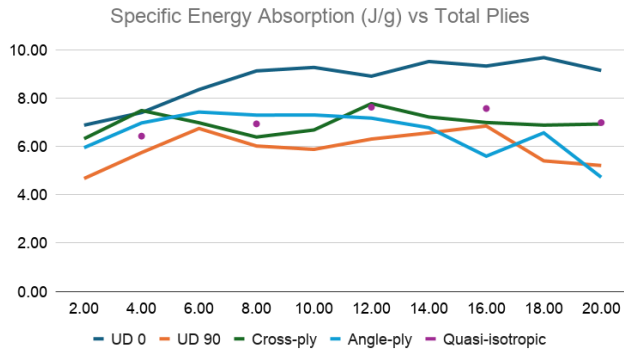


Figure 11: Specific Energy Absorption (J/g) vs Total Plies

In Figure 12, the internal energy absorption of the composites is plotted as a ratio against that of the steel control armor. A value of 1.00 would represent an equal performance in energy absorption. As shown in Figure 12, the energy absorbed in the composite is lower than that of AR500 steel armor for all layups, but increases significantly with the number of plies. When considered that the mass ratio increases linearly with the number of plies from roughly 1.5% to 15%, it becomes clear that energy absorption ratio compared to steel increases at a faster rate than the mass ratio, for the CFRP laminates.

7.3 Low Velocity Beam Impact Simulation Results

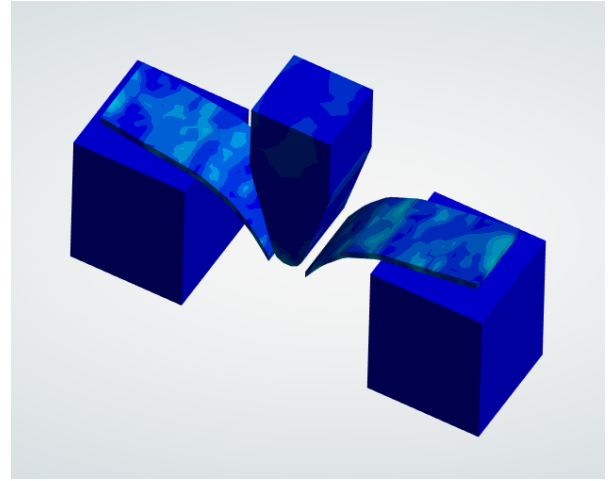


Figure 13: Unidirectional $[90]_8$ ply Stackup

Figure 14 shows the total energy absorbed for the different laminate stack-ups as the number of plies increased. In general, increasing the number of plies increased the total energy absorbed for almost all laminate configurations. The UD 0 laminate consistently absorbed the most energy before failure, while the UD 90 laminate absorbed the least. In the UD 90 laminate, the fibers were aligned with the edge of the impactor, causing more of the bending load to be carried through the weaker matrix-dominated direction of the laminate. This resulted in earlier localized failure and much lower energy absorption. In contrast, the UD 0 laminate oriented the fibers perpendicular to the impactor edge, allowing more of the impact load to be carried through the fibers themselves, resulting in greater energy absorption before failure.

The cross-ply, angle-ply, and quasi-isotropic laminates generally fell between the two extremes. Cross-ply laminates showed relatively high energy absorption at lower ply counts, though the increase became smaller as plies were added. The angle-ply laminate showed a more grad-

ual increase in absorbed energy as the number of plies increased. Similar trends can be seen in Figure 15, which shows the specific energy absorption for the different laminates. While the total absorbed energy generally increased with ply count, the specific energy absorption remained relatively constant or slightly decreased for several layups.

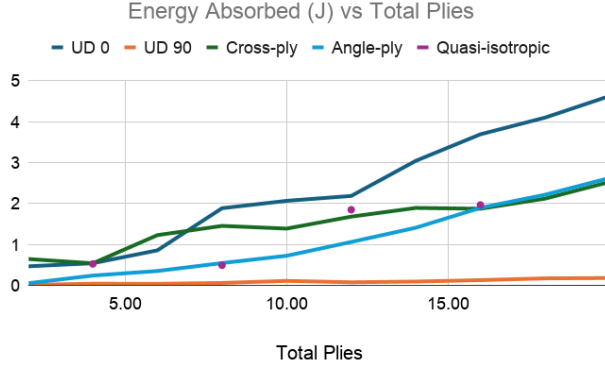


Figure 14: Energy Absorbed (J) vs Total Plies

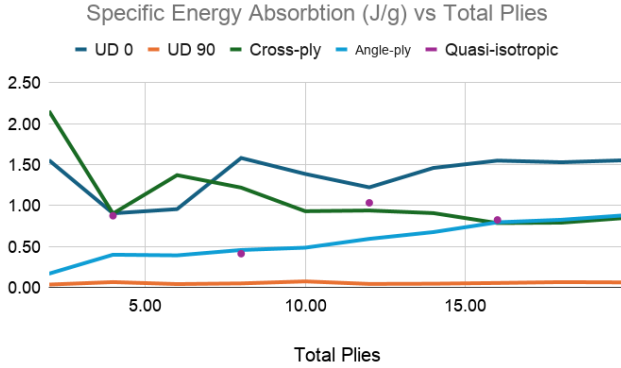


Figure 15: Specific Energy Absorption (J/g) vs Total Plies

7.4 Limitations

There are several limitations with the results that we have obtained, and we recognize that our results are only a first cut at analyzing CFRP composites for combat robotics applications.

One limitation of our setup is that the boundary conditions of our armor was simply to fix the left and right edges of the plate, when this is not actually how armor is mounted in combat robotics. Instead, fixtures for armor are typically distributed throughout the area of the plate with fasteners, and these fastener holes could induce stress concentrations and failure modes that are not represented in our setup.

An assumption was made that the impactor was a rigid body; in reality, it would also experience deformation,

which can impact the amount of energy transfer and the effectiveness of further impacts. This is also one of the main reasons we could not design an experimental setup to validate our simulations, since the impactor would be affected after each impact. Though we could theoretically account for the error that would arise from that, it was decided to not be worth the time and budget to ensure the safety of a hypothetical experimental setup.

Another limitation came with the laminate setup. The entire laminate was created using only one surface, which wouldn't capture certain failure modes such as delamination. This was done to limit computational runtime.

Another limitation is with the parameters in the MAT054 model. The parameters have not been tuned based on experimental data, which Cherniaev has shown is necessary for stronger correlation with experimental data, at least for progressive axial crush tests [15]. It is generally better to tune model parameters based on a similar experimental setup, but since we did not have an experiment with real-world data, the results from our model with default parameter values likely has slightly skewed data.

A final major limitation which has already been mentioned is that our simulation erodes elements that have reached a strain limit, meaning that the internal energy absorbed by those deleted elements is not captured by the simulation.

8 Conclusion

The objective of this project was to evaluate and determine the lightweight alternative of our CFRP laminate armor to AR500 steel weapon for impact protection. As AR500 steel was a common use for the control material in combat robotics armor, we use the CFRP laminates configuration from our simulations and compared them under LS-DYNA impact conditions.

As a result, increasing the number plies leads to an increase of maximum internal energy absorbed by the laminate. The cross-ply and angle-ply configurations showed slightly stronger energy absorption trends than other layups, but none of the laminates fully matched the total absorbed energy of the AR500 steel control. However, CFRP laminates had much lower mass, which still showed their efficient potential as lightweight materials. This also shows how deformation would not be a major concern for the impact of armor. Although the composite plates generally experienced higher deformation than AR500, which could be problematic, as this is due to limitations of the simulation.

Overall, the simulations suggest that CFRP laminates may not fully replace AR500 steel for impact resistance. However, they can provide useful weight savings and energy absorption for mass-constrained applications in combat robotics.

Future areas of interest include impacts on angled ar-

mor, hybrid material stacks, a more comprehensive simulation setup, and validation of the results on a real robot design; most of these are meant to address our current limitations.

For potential integration of the robot armor layout, alloys and other isotropic materials can be part of a future hybrid-stackup strategy rather than utilizing standalone composite plates. Titanium or aluminum alloys may be used in the composite stackup, particularly on the outer surface, forming fiber-metal laminates that can resist initial contact and indentation from the weapon impact. This also helps CFRP plies absorb more energy by increasing fracture and shear damage.

An example that indicates this is an experiment with a 7.62mm projectile against an aluminum 7075 composite armor layer. Its ballistic performance had very low deformation and partial penetration due to the sandwich inter-layer structure of the composite armor, with a lightweight core that connects to the solid face sheets, providing high resistance to bending and buckling, excellent shear stiffness, and high energy absorption capabilities [16]. As such, we are to believe that simulating a similar composite configuration would yield better results.

9 Author Contributions

Nathan Lam led and organized the project, created the original abstract and background, designed the initial simulation setup, simulated and recorded data for the control simulation, and contributed to the report where necessary.

Nathaniel Nunez developed the finalized plate impactor simulation base, learned ACP and Ansys Mechanical parameterization implementation allowing data collection for all CFRP laminate configurations, troubleshoot material props for accurate impact sims.

Ryan Chiang wrote the abstract, introduction, background, objective, methods, revised and edited the Ansys LS-DYNA Setup, assisted in creating the graphs in Google Sheets for the tables in Results and Analysis.

Micah Hsu revised the Methods section, drafted part of the Results section, and contributed to the report in general.

Ernesto Tellez Perez developed and finalized Charpy-inspired beam impact simulation and CAD, implemented ACP // Mechanical parameterization for laminate configurations, performed data collection and analysis, and troubleshoot composite material and failure properties for impact simulations.

Bhavya Mamnani performed literature research and revised conclusion.

Khoi Nguyen conducted research and wrote the Conclusion of the result.

10 Generative AI Tools

Generative AI was used minimally, with it used primarily for troubleshooting issues with our simulation setup to reduce the need of looking through big user manuals. Generative AI was also utilized to assist in formatting the Latex file and references.

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